

Embedded Summer Internship Training Program Syllabus

S.No.	Day Order	Theory/LAB	Lab
1	Day-1	Embedded Systems & ARM Architecture: Introduction to Embedded Systems-Applications of Embedded Systems- Embedded System vs General Purpose System- Microcontroller role in industry-ARM Design Philosophy RISC vs CISC- ARM Programmer's Model	Study of various Microcontroller and Processor Development Boards Study ARM Cortex-M architecture block diagram Observe memory map and registers
2	Day-2	Basic Electronics & Embedded C Programming: Basic Electronics (Resistors, Capacitors, Diodes, Transistors, Voltage Regulators)- Voltage Regulators- Embedded C Programming- Data Types, Operators- Control Statements and Loops Arrays, Functions, Pointers, Structures- Bitwise Operations	Transistor switching demonstration C programs using loops and functions Array and pointer programs
3	Day-3	STM32 Architecture & Tools: STM32 Family Overview- STM32 Architecture- Pin Description- Memory Map -Clock Configuration- STM32CubeIDE and STM32CubeMX	Install STM32CubeIDE Create new project using CubeMX Configure system clock Generate and build project
4	Day-4	GPIO & Interrupt Programming: GPIO Programming using HAL- LED Blinking- Push Button Input- External Interrupt Programming	LED blinking using HAL Push button interfacing Toggle LED using external interrupt Multiple LED control using GPIO
5	Day-5	Timers, ADC, DAC & Communication: Timer Programming- PWM Generation- ADC Programming- DAC Programming UART, SPI, I2C Overview	PWM generation for LED brightness Read analog sensor using ADC UART communication between PC and STM32
6	Day-6	HAL vs Bare Metal: HAL vs Bare Metal Programming - Register Level Programming-ARM Cortex-M Memory Map	Blink LED using register programming Enable GPIO peripheral clock using RCC register Access memory mapped registers
7	Day-7	GPIO & RCC Register Programming: GPIO Registers-RCC Registers-Peripheral Clock Enable-GPIO Configuration Registers	Configure GPIO pin using registers Enable peripheral clock using RCC registers Register-level LED blinking program

8	Day-8	Interrupt, Timer & ADC Registers: External Interrupt Registers-Timer Registers-ADC Registers	External interrupt using registers Timer delay using registers ADC read using register programming
9	Day-9	Communication Protocols: UART Register Programming-SPI Register Programming-I2C Register Programming	Exercise with UART,SPI,I2C Protocols
10	Day-10	PWM & Driver Development: PWM using Registers-Driver Development Concepts-Driver Architecture	PWM generation using timer registers Write simple GPIO driver
11	Day-11	Driver Implementation: GPIO Driver Development-SysTick Timer Driver-Timer Driver-UART Driver	Implement GPIO driver functions Implement SysTick delay driver Implement UART driver
12	Day-12	IoT Basics: Introduction to IoT-IoT Architecture-Sensors and Actuators-ESP32 / CC3200-IoT Communication Models-HTTP Protocol	Send sensor data using HTTP
13	Day-13	Cloud & Communication: MQTT Protocol-Cloud Platforms-IoT Dashboard	Send sensor data to cloud using MQTT Create IoT dashboard Visualize sensor data
14	Day-14	Machine Learning & Edge AI: Introduction to Machine Learning-TinyML Concepts-Edge AI-Edge Impulse Platform-Model Training and Deployment	Collect sensor data Train model using Edge Impulse Deploy model to microcontroller
15	Day-15	Summary, Project Discussion & Assessment: Summary of Embedded Systems Concepts- -Summary of STM32 HAL and Bare Metal - Programming-Communication Protocol Review-IoT and TinyML Overview-Project Design Discussion- Industry Applications Discussion	Mini Project Project Discussion Assessment / Quiz Program Feedback

**** Note : Open-source tools will be used for the training. If any relevant licensed tools are available in the premises, they may also be utilized.**